Open-Source versus Proprietary LLMs for Sentiment-Based Equity Portfolios

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Abstract

This study evaluates seven Large Language Models, proprietary (GPT, Claude, Gemini) and open-source (Llama, Mistral), on extracting trading signals from over 1.4 million headlines, using rank-based portfolios of S&P 500 constituents, 2021–2025. Portfolios from the larger models generate positive alpha; the smallest does not. Proprietary models produced the highest risk-adjusted returns, with Claude 3 Haiku reaching a Sharpe Ratio of 0.94; its successor Claude 4.5 Haiku ranked second, so the result does not rest on one now-deprecated model. The open-source Llama 3.1 8B performed close to commercial APIs, a usable local alternative subject to cost and privacy trade-offs.

Keywords: Large Language Models, Sentiment Analysis, Quantitative Trading, Portfolio Management, Natural Language Processing

1. Introduction

Large Language Models (LLMs) now have direct applications in finance, including investment strategies and sentiment analysis. The use of textual data as a trading signal goes back to Tetlock (2007). Earlier work relied on dictionary methods, notably the word lists of Loughran and McDonald (2011), which lack contextual awareness and often miss the meaning of financial narratives that depend on surrounding text.

Domain-specific models such as FinBERT (Araci, 2019) and BloombergGPT (Wu et al., 2023) improved on generic sentiment analysis, and pre-trained transformers have been used to build sentiment indicators from text (Seiler, 2025). Several studies report LLM outperformance in return forecasting under specific conditions (Lopez-Lira and Tang, 2024; Pelster and Val, 2024; Kirtac and Germano, 2024; Liu

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and Shi, 2025), while others are cautious: Perlin et al. (2025) find no consistent edge over naive portfolios, and Leippold (2023) warns of sentiment manipulation.

Most of this work tests single proprietary models, often the OpenAI GPT series (Pelster and Val, 2024; Romanko et al., 2023; Schneider and Yilmaz, 2025) or Google's Gemini (Perlin et al., 2025), with a recent strand calling for multi-model evaluation to capture output variability (Lopez-Lira and Tang, 2024; Kirtac and Germano, 2024; Chon et al., 2025; Didisheim et al., 2025; Moreno and Ordieres-Meré, 2025). Less examined is how commercial models compare with open-source ones that can be run locally on the same task. Proprietary models run as paid "black boxes" with undisclosed parameter counts (Brown et al., 2020; Hoffmann et al., 2022), while open-source models such as LLaMA (Touvron et al., 2023a) and Mistral (Jiang et al., 2023) can be deployed and inspected locally.

This paper compares proprietary (GPT, Claude, Gemini) and open-source (Llama, Mistral) models in equity portfolio management. Unlike studies that use a small asset subset or direct stock picking, the analysis covers the full set of S&P 500 constituents, classifying each headline as positive, negative, or neutral and aggregating these into monthly sentiment scores; the raw extraction exceeds 1.6 million headlines (2021–2025), of which 1,477,385 remain after cleaning (Table 1). Portfolios are formed with a rank-based rule that uses relative signal strength rather than absolute scores, limiting the effect of uneven news coverage. The aim is to show how the choice between commercial and locally deployed models shifts the trading signal, with excess returns over the S&P 500 Equal Weighted (S&P 500 EW) that vary by model size and type, not to benchmark every model.

2. Data and Methodology

The dataset was extracted from the EOD Historical Data portal[1] and covers headlines for United States listed companies. The scope was limited to S&P 500 companies from January 2021 to December 2025; earlier years were excluded because headline volume was too low for reliable analysis. Index membership was treated dynamically: each month included only the firms that were S&P 500 constituents that month, using the historical membership rather than the final 2025 list, which avoids selecting firms on later index inclusion or survival.

Cleaning then removed metadata entries and duplicate headlines. To remove near-duplicates, a normalized Levenshtein similarity ratio (Levenshtein, 1966) was applied, and pairs scoring above 0.6 for the same company on the same date were

[1] https://eodhd.com/

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dropped (Lopez-Lira and Tang, 2024). Restricting the comparison to same-day, same-firm headlines limits the chance of removing economically distinct news.

Table 1 presents descriptive statistics after cleaning, highlighting two features. First, coverage is heavily skewed toward a subset of large-cap firms, though the long tail still covers the remaining constituents. Second, volume more than doubles in 2025: the vendor began extracting headlines from unstructured sources with a large language model[2], without a published start date. The full dataset was retained, since the Sentiment Index is normalized within each firm-month and the volume change does not mechanically shift the signal.

Table 1: Descriptive Statistics of Financial News Headlines (2021–2025)

Metric	**2021**	**2022**	**2023**	**2024**	**2025**	**Total**
Panel A: Data Filtering Pipeline						
Raw Headlines Extracted	246,365	281,361	291,426	230,335	640,082	1,689,569
Removed: Metadata Artifacts	5	18	1,997	6,406	0	8,426
Removed: Near-Duplicates[a]	41,657	21,757	28,952	19,899	91,493	203,758
Final Analytical Sample	**204,703**	**259,586**	**260,477**	**204,030**	**548,589**	**1,477,385**
Panel B: Temporal Distribution (Monthly)						
Mean Monthly Volume	15,405.8	20,204.3	20,247.2	15,291.2	44,981.3	23,226.0
Std. Dev. Monthly Volume	6,702.3	3,004.5	1,966.5	5,191.7	25,682.7	16,246.5
Panel C: Cross-Sectional Coverage (Tickers)						
Mean Headlines per Firm	400.2	515.9	527.0	382.3	1,054.2	2,538.4
Std. Dev. per Firm	783.9	986.0	983.8	671.9	2,569.9	5,363.0
Concentration (Top 10 Firms)	25.7%	24.3%	24.9%	22.6%	31.2%	25.7%

[a] Similarity threshold set at $\delta > 0.6$ using a normalized Levenshtein similarity ratio.

A standardized prompt based on Lopez-Lira and Tang (2024) and Didisheim et al. (2025) was applied to the entire dataset, so the labels follow the same definition of news impact on investor expectations used in those studies; testing alternative designs is left to future work. For efficiency, headlines were batched into queries of up to 10 items each. The prompt was structured as follows:

[2]https://eodhd.com/financial-apis/stock-market-financial-news-api

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```
Forget all your previous instructions. You are a financial expert with
stock recommendation experience. Your task is to classify financial
headlines.
For each headline, respond with "YES" for good news, "NO" for bad
news, or "UNKNOWN" for uncertain news. You must classify each headline
individually.
Your final output must be a single, valid JSON object and nothing
else. Do not include any introductory text, explanations, or markdown
formatting like "json". The JSON object must map the headline number (as
a string key) to its classification.

Example for 3 headlines:
{"1": "YES", "2": "UNKNOWN", "3": "NO"}
```

2.1. LLM Selection and Classification

Given the dataset's scale, cost-efficient models were prioritized over computationally intensive ones, supported by evidence that reasoning-enhanced models do not necessarily outperform standard ones in financial text tasks (Didisheim et al., 2025). The models fall into two groups: efficient proprietary models (Flash, Haiku, Nano) and open-source models (Llama, Mistral) restricted to the 1B and 7–8B parameter ranges for local deployment (Touvron et al., 2023b). Claude 3 Haiku is being deprecated by its provider, so its successor, Claude 4.5 Haiku, was also run to test whether a newer version of the same family changes the result. Table 2 summarizes these specifications.

Table 2: Technical Specifications of Selected Large Language Models

Developer	**Model**	**Params**	**Context**	**Data Cutoff**	**Access Date**
Panel A: Proprietary Models (Closed-Source)					
OpenAI	GPT-4.1 Nano	Undisclosed	1M	Jun 2024	Jun 2026
Anthropic	Claude 3 Haiku	Undisclosed	200k	Aug 2023	Sep 2025–Feb 2026
Anthropic	Claude 4.5 Haiku	Undisclosed	200k	Feb 2025	Jun 2026
Google	Gemini 2.5 Flash	Undisclosed	1M	Jan 2025	Sep 2025–Feb 2026
Panel B: Open-Source Models (Weights Available)					
Meta	LLaMA 3.1 8B Instruct	8 Billion	128k	Dec 2023	Oct 2025–Jan 2026
Meta	LLaMA 3.2 1B Instruct	1 Billion	128k	Dec 2023	Jan–Feb 2026
Mistral AI	Mistral 7B Instruct v0.2	7 Billion	32k	Dec 2023[a]	Sep 2025–Jan 2026

Note: "Context" refers to the maximum number of input tokens the model can process. "Access Date" gives the months over which each proprietary model was queried via API and each open-source model was run locally. [a]Mistral AI does not officially disclose a strict knowledge cutoff; the model's release date is used as a proxy.

For reproducibility, decoding was deterministic: the proprietary models were queried through their providers' APIs at temperature zero, and the open-source

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models used greedy decoding (do_sample=False) (Brown et al., 2020). The open-source models were run locally in half precision (FP16) using the Hugging Face transformers library (Wolf et al., 2020). The proprietary models were accessed on the dates listed in Table 2, as commercial APIs can change between releases.

2.2. Portfolio Selection

To translate the LLM classifications into an investment signal, a normalized Sentiment Index (SI) was calculated for each company i at month t:

$$\mathrm{SI}_{i,t} = \frac{\sum \mathrm{POS} - \sum \mathrm{NEG}}{\sum \mathrm{POS} + \sum \mathrm{NEG}}, \tag{1}$$

where $\sum \mathrm{POS}$ and $\sum \mathrm{NEG}$ denote the total count of positive and negative headlines, respectively. Mirroring established polarity measures (Loughran and McDonald, 2011), this metric captures sentiment intensity on a $[-1, 1]$ scale, independent of coverage volume.

Portfolios were constructed at the beginning of each month $t + 1$ from signals aggregated during month t. Stocks were ranked by $SI_{i,t}$ in descending order, with a minimum target of 15 assets to mitigate unsystematic risk (Elton and Gruber, 1977). A flexible tie-breaking rule included all stocks at the threshold score (Jegadeesh and Titman, 1993), with allocations equal-weighted. A minimum of 15 headlines per firm per month was also required, restricting the portfolio to firms with sustained media coverage, in line with coverage filters in the literature (Fang and Peress, 2009). A side effect is that the eligible universe tilted toward larger, more visible firms, so the strategy reflected a sentiment signal within a media-covered subset rather than across all S&P 500 names.

Each basket was held for one month and rebalanced on the first business day of month $t + 1$. Returns were computed from close prices adjusted for splits and dividends, and therefore represent total returns. The benchmarks were the S&P 500 EW, an equal-weighted portfolio of the active index members each month, and a Random Portfolio of 15 stocks drawn from those same members, averaged over 1,000 Monte Carlo draws. Both drew on the point-in-time membership rather than the headline-eligible subset.

Annualized return, volatility, and the Sharpe Ratio follow the standard monthly-to-annual scaling, with a zero risk-free rate. Alpha and beta were estimated from an OLS single-factor regression of each strategy's monthly return on the monthly S&P 500 EW return; beta is the slope and alpha the intercept, annualized by compounding.

Two steps limited look-ahead bias: headlines were stripped of dates and processed

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in temporal-agnostic batches, and the models classified text without access to asset prices (Vaswani et al., 2017; Brown et al., 2020). These steps do not fully remove the risk that a model saw an event during pre-training, since the cutoffs in Table 2 overlap part of the sample. The analysis was therefore repeated after each model's cutoff: apart from the two most recent models, whose post-cutoff window is too short, the alpha kept its full-sample sign.[3]

3. Results

Table 3: Annualized Performance Metrics (2021-2025)

Strategy	Ret (%)	Vol (%)	Sharpe	Max DD (%)	Alpha (%)	Beta
Panel A: Proprietary Models						
Claude 3 Haiku	**14.18**	15.11	**0.94**	-17.14	**7.73**	0.90
Claude 4.5 Haiku	12.61	**14.34**	0.88	-18.49	6.66	0.82
Gemini 2.5 Flash	11.25	14.91	0.75	**-16.28**	4.73	0.89
GPT-4.1 Nano	10.41	15.87	0.66	-23.20	3.82	0.89
Panel B: Open-Source Models						
Llama 3.1 8B	11.53	15.50	0.74	-20.34	4.77	0.92
Mistral 7B	7.68	15.00	0.51	-20.39	0.79	0.92
Llama 3.2 1B	4.61	15.44	0.30	-23.59	-2.18	0.91
Panel C: Benchmarks						
S&P 500 EW	7.46	15.67	0.48	-21.45	0.00	1.00
Random 15 (Avg)	7.42	15.69	0.47	-21.58	-0.05	1.00

Note: Performance metrics are annualized. "Ret" denotes Annualized Return, "Vol" denotes Annualized Volatility, and "Max DD" represents the Maximum Drawdown over the period. The benchmarks and portfolios are calculated from close prices adjusted for both splits and dividends, and therefore represent total returns.

Table 3 reports the annualized risk and return metrics for the full sample period (January 2021 to December 2025). The excess returns are smaller than the large figures reported in early unconstrained LLM strategies (Lopez-Lira and Tang, 2024; Kirtac and Germano, 2024), which is consistent with the rank-based portfolio construction used here.

In this sample, the strongest strategies were proprietary, though the open-source Llama 3.1 8B also beat the benchmark. As shown in Figure 1, the cumulative wealth of the top strategy (Claude 3 Haiku) separates from the S&P 500 EW in early 2022, and the gap widens during 2024 and 2025. Part of this later separation coincides with the strong performance of large technology firms in 2024–2025, so the period itself may contribute to the measured outperformance.

[3]Full out-of-sample results are available from the authors on request.

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Claude 3 Haiku had the highest risk-adjusted return, about double the Sharpe Ratio of the S&P 500 EW benchmark (Table 3). Its newer successor, Claude 4.5 Haiku, ranked second, with the lowest volatility and beta in the sample but a lower alpha than Claude 3 Haiku. Gemini 2.5 Flash and GPT-4.1 Nano also returned more than the market. Given the monthly frequency over 2021–2025, these differences are reported as descriptive rather than as formally tested differences in alpha or Sharpe Ratio.

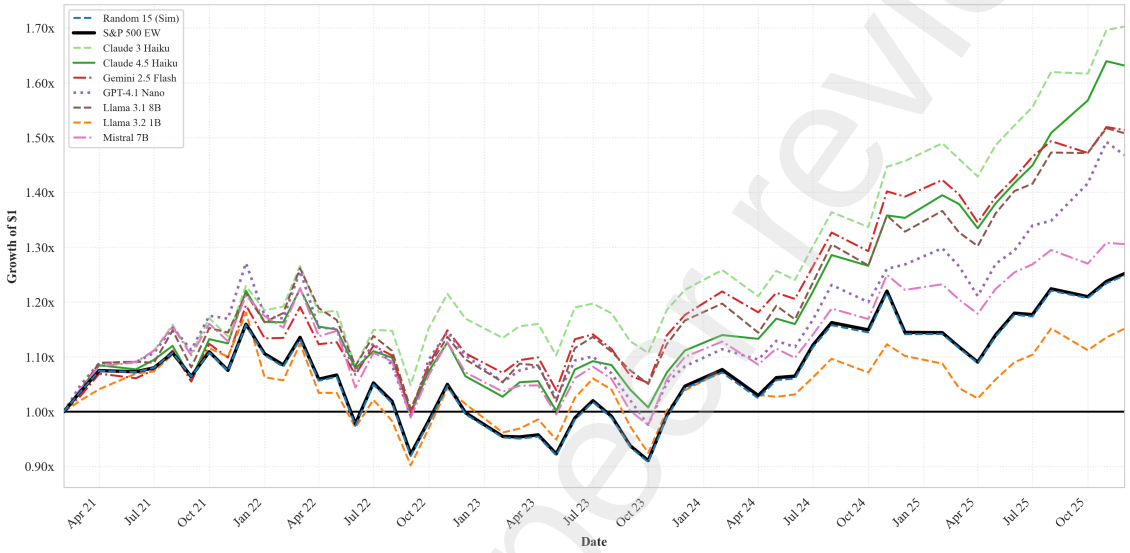


Figure 1: Cumulative Wealth of LLM Strategies vs. Benchmarks (2021–2025). Note the divergence of the leading strategies (top lines) during the 2022 downturn and the underperformance of the 1B parameter model (dashed orange line).

The open-source results depend on more than parameter count. Llama 3.1 8B ranked close to Gemini 2.5 Flash and ahead of GPT-4.1 Nano, though below the two Haiku models. Mistral 7B, with a similar parameter count, finished only slightly above the benchmark. The gap between Llama 8B and Mistral 7B suggests that within the 7–8B range, training data and instruction-following matter alongside size, though with only two models in this range the comparison is limited. Moreno and Ordieres-Meré (2025) report a related pattern, with a smaller finance-tuned model outperforming a larger general one on analyst-report sentiment.

The smaller Llama 3.2 1B performed worse. It did not outperform the benchmark, with a return of 4.61% and a negative alpha of -2.18%, and it lags the S&P 500 EW for most of the sample (orange dashed line in Figure 1). This is consistent with the 1B model being less able to separate noise from informative sentiment, but with a single 1B model in the sample the result could also reflect its instruction-following or training data rather than size alone. It is therefore evidence on this specific model rather than a general size threshold.

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Several portfolios also showed smaller drawdowns than the benchmark. During the 2022 bear market, Claude 3 Haiku, Claude 4.5 Haiku, and Gemini 2.5 Flash limited losses below the S&P 500 EW, while GPT-4.1 Nano was an exception with a deeper drawdown (Table 3). The beta coefficients, near or below 0.90, indicate that these portfolios stayed correlated with the market but, for the stronger models, with somewhat lower downside exposure.

The excess returns come mainly from the alpha term rather than from market exposure. Claude 3 Haiku had the highest annualized single-factor alpha at 7.73%, and the open-source Llama 3.1 8B reached 4.77%, above the 3.82% of GPT-4.1 Nano. This alpha is measured against the market benchmark only and is not adjusted for size, value, momentum, or sector exposure, so part of it may reflect such exposures rather than sentiment alone. With that caveat, the positive alpha from the Llama 3.1 8B model is an encouraging result for locally deployed open-source models.

4. Conclusion

Large Language Models can turn financial headlines into sentiment-based trading signals, and in this sample the resulting portfolios outperformed both the equal-weighted benchmark and a random selection. By normalizing sentiment scores and ranking on relative signal strength, the rank-based construction managed the changing volume of news, and the larger models produced positive alpha while the smallest did not. These results hold for the data, market, period, prompt, and portfolio rule studied here, and are best read as evidence from this setting rather than a general claim.

The proprietary models, in particular Claude 3 Haiku, gave the highest risk-adjusted returns, while the open-source Llama 3.1 8B performed close to them and ahead of GPT-4.1 Nano. A locally deployable open-source model is therefore a usable alternative for researchers who would otherwise depend on commercial APIs that are deprecated quickly. That concern is concrete: Claude 3 Haiku is being deprecated, yet its successor Claude 4.5 Haiku ranked below it, so a newer release does not guarantee a stronger signal. The 1B model did not beat the market. Cost, latency, and hardware are discussed rather than measured, leaving users to weigh the convenience of commercial APIs against the data privacy and cost control of a local model.

Future work could extend the analysis in three directions. This study omits trading costs, so adding transaction frictions and fees would test net profitability. Comparison with domain-specific encoders such as FinBERT would contrast generative and discriminative classification. Finally, the design reduces temporal

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look-ahead bias but not entity-level effects from pre-training; anonymizing company names could help separate semantic interpretation from learned brand associations.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author(s) used Google Gemini in order to improve readability of the final text. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

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